Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science — West Lafayette, IN
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics — May 2028
GPA 3.32/4 | Clubs: Club Swimming, House Band

PROFESSIONAL EXPERIENCE

Boston Dynamics — Waltham, MA
Sensors and Perception Co-Op, Atlas — Jul 2026 – Present

- Gave Atlas test engineers a view of ~28k impact events from ~900 robot test logs by building an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM, with telemetry ETL into SQLite, a 3D impact-location heatmap, and per-event waveforms.
- Predicting steady-state temperatures across hundreds of units without full-length tests by fitting heating curves with multi-start least squares in Python (Pandas, SciPy).
- Computing limb velocity and energy at impact with floating-base forward kinematics from joint encoders and Khatib's operational-space effective mass, mapping impacts onto the robot's mesh, and building a MuJoCo impact simulation.

Caterpillar (via Purdue Data Mine) — West Lafayette, IN
Machine Learning Researcher — Jan 2026 – May 2026

- Cut paint-line material usage 4.7% with LSTM and XGBoost models forecasting next-day primer, topcoat, and solvent consumption from five years of data, each forecast carrying a Monte Carlo confidence interval.
- Contributed to a 3.3% reduction in manufacturing cycle time by building Python pipelines that consolidated five years of noisy legacy data into one daily series.

Chortos Lab, Soft Adaptive Systems and Structures — West Lafayette, IN
Undergraduate Researcher — Aug 2025 – May 2026

- Increased gel force output by over 50% across formulation rounds, measured with a Python computer-vision tracker I wrote that resolves displacement to 0.01 mm, plus LabVIEW/MATLAB acquisition of 50,000+ point datasets.

PROJECTS

Legged Robot State Estimation and Control | *Personal Project* — Aug 2026 – Oct 2026

- Reached 0.013 m/s velocity RMSE over a 60 s walk, 100× lower than IMU-only dead reckoning (1.38 m/s), with a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU and leg kinematics on a simulated Unitree Go1, at ~20 µs per update.
- 0 falls in 5 × 60 s walks with the walking controller running on the estimate, tracking within 0.004 m/s of the ground-truth-fed run; a 500 Hz QP force-balance controller then cut ground-truth tracking error 46% (0.169 to 0.091 m/s).
- Removed a 4° lean by tracing it to weak tilt observability on straight walks and adding a 4 s startup turn; Jacobians verified against finite differences in GoogleTest.

Active Orthosis, Purdue ASME Biomechanics | *Electronics Team Lead* — Apr 2024 – May 2026

- Cut EMG-to-motion latency 58% (103 ms to 43 ms) with a Bayesian EMG estimator and a 20 ms onset confirm that took false starts to zero, in closed-loop simulation on real recorded EMG, pinned by 26 GoogleTest tests in CI.
- Held speed error to 0.8% (vs. 98% feed-forward alone) lifting the forearm in simulation with a 1 kHz PI velocity controller in C++ on an ESP32-S3, protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Raised cross-day gesture decoding from 31% to 87% in emulation over 115 days of real EMG by leading a 5-engineer team and redesigning the 4-channel EMG board in KiCad (INA333, right-leg drive, ISO7763 isolation).

Asteroid Hopping Robot | *Independent Research, Purdue University & UC3M (Madrid)* — Oct 2026 – Present

- Designing push-off control optimized to hit a target launch speed, knee impedance control for landing, and reaction-wheel reorientation for a three-legged hopping robot, simulated in MuJoCo to trade hop distance against escape and tip-over risk.

SKILLS

- **Programming:** C++17, Python, MATLAB, SQL | Languages: English, Turkish (fluent), Spanish (fluent)
- **Robotics and controls:** Kalman filtering (error-state EKF), sensor fusion, rigid-body kinematics, PI and feed-forward control, QP force control, MuJoCo, Eigen, FreeRTOS, ESP32
- **Software and data:** Linux, Git, GitHub Actions CI, CMake, GoogleTest, Flask, Plotly, SQLite, Pandas, SciPy, PyTorch, XGBoost, Snowflake
- **Hardware and CAD:** KiCad, LabVIEW, DAQ, CATIA, NX, Fusion 360, Creo, FEA